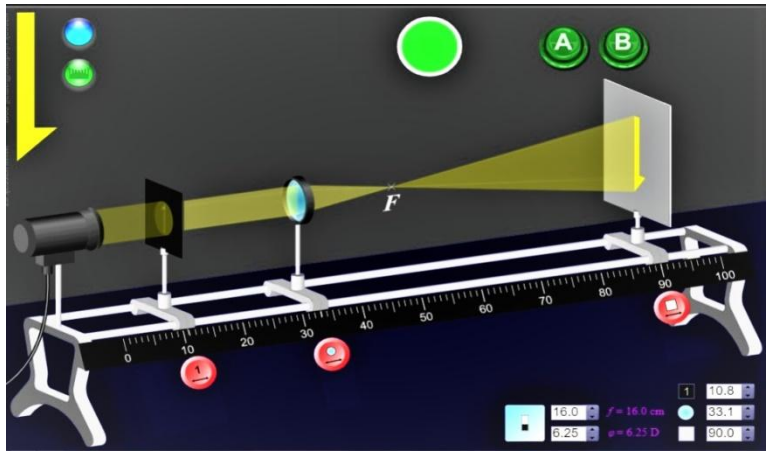
# Fizika

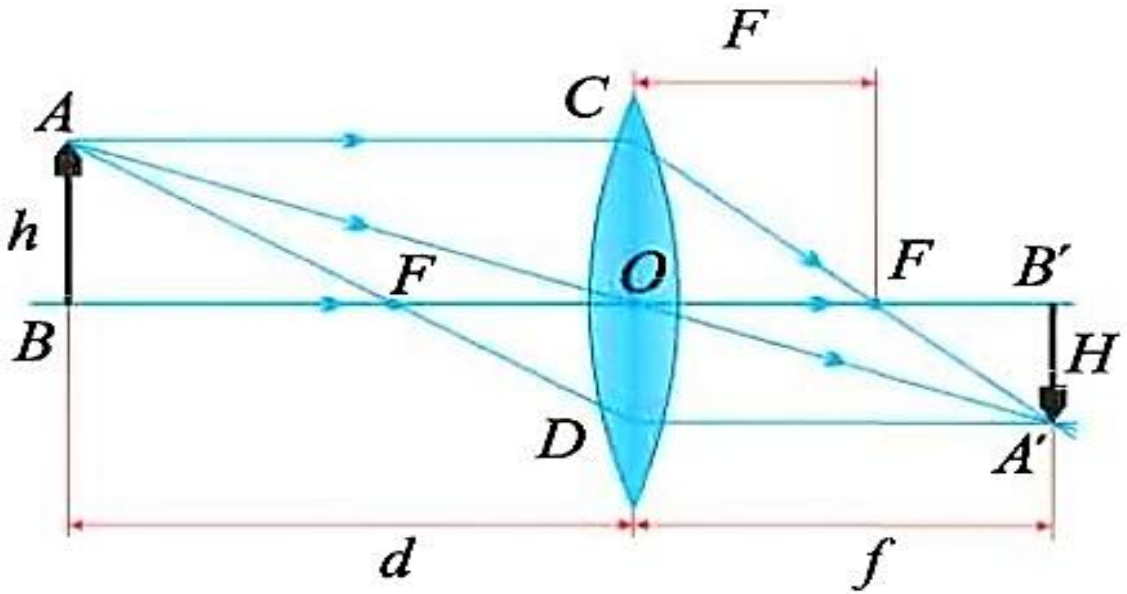
9-sinf

## Mavzu:

# Yupqa linza yordamida tasvir yasash



# Yig‘uvchi linzada tasvir yasash




$AB$ buyum $A'B'$ buymning tasviri.

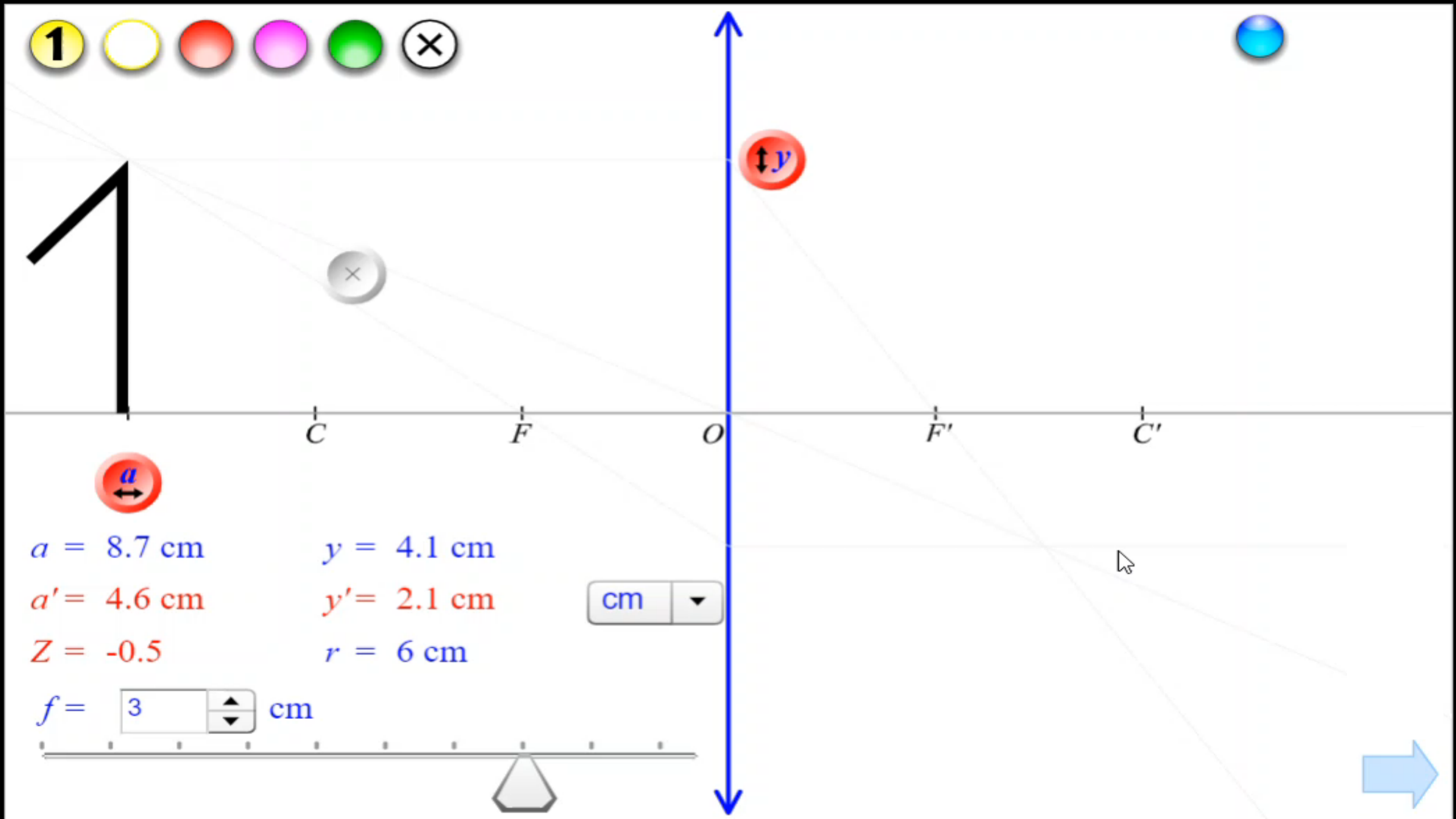
# Linza formulasi va chiziqli kattalashtirishi

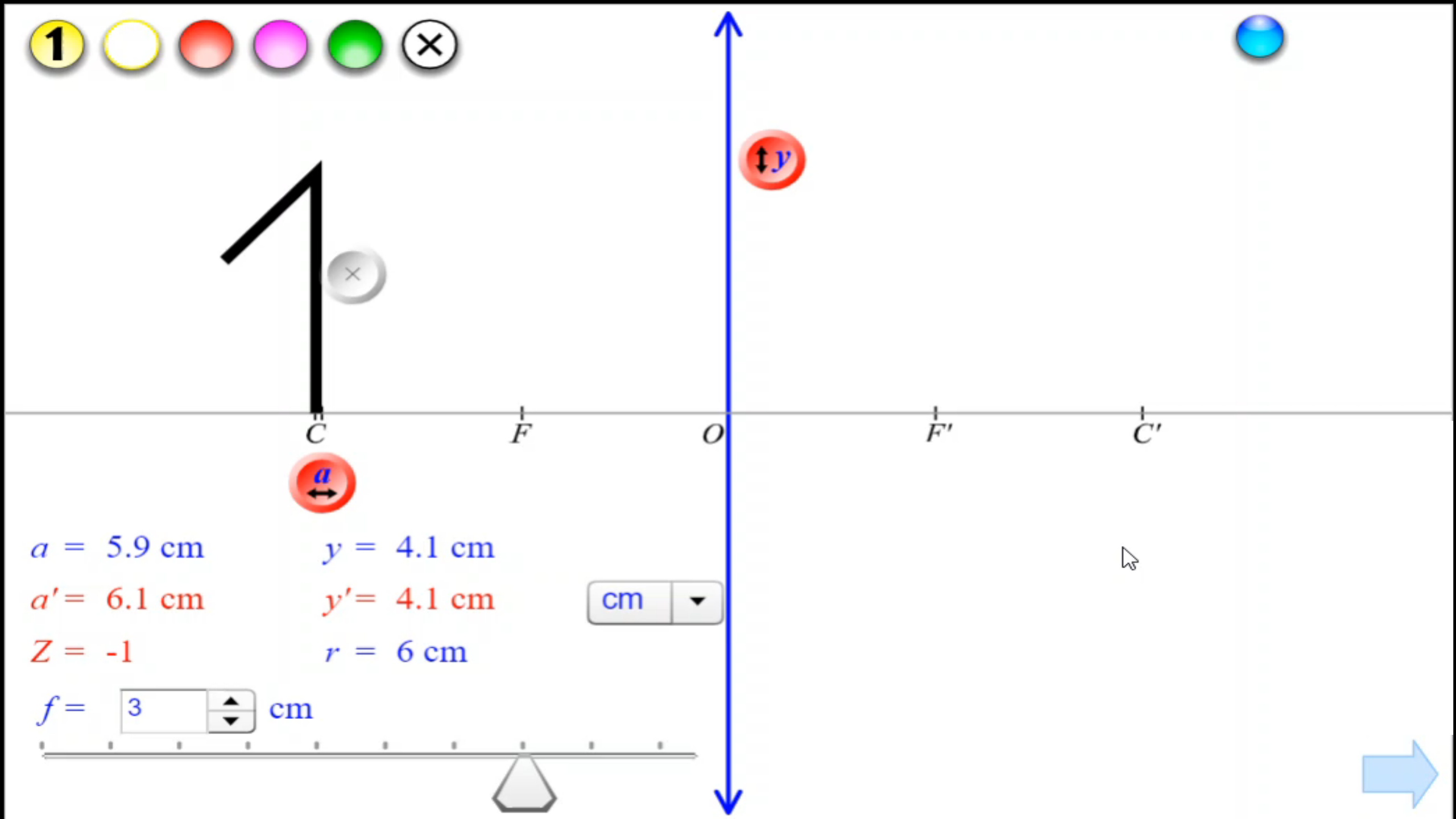
Linza formulasi buyumdan linzagacha bo‘lgan $d$ masofa, linzadan tasvirgacha $f$ masofa va linzaning $F$ fokus masofasi orasidagi bog‘lanishni ifodalaydi.

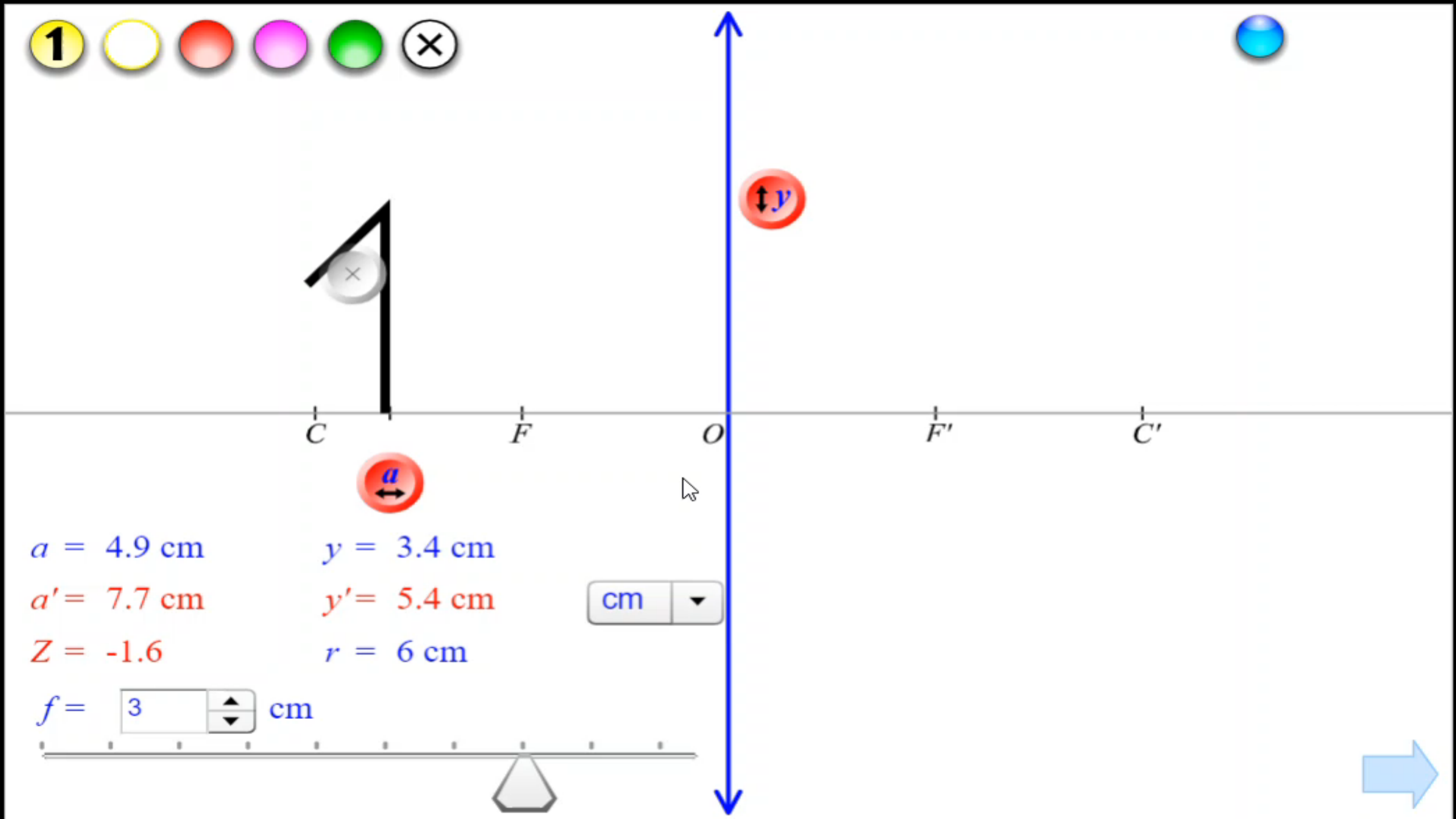
$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f} \qquad \text{yoki} \qquad D = \frac{1}{d} + \frac{1}{f}$$

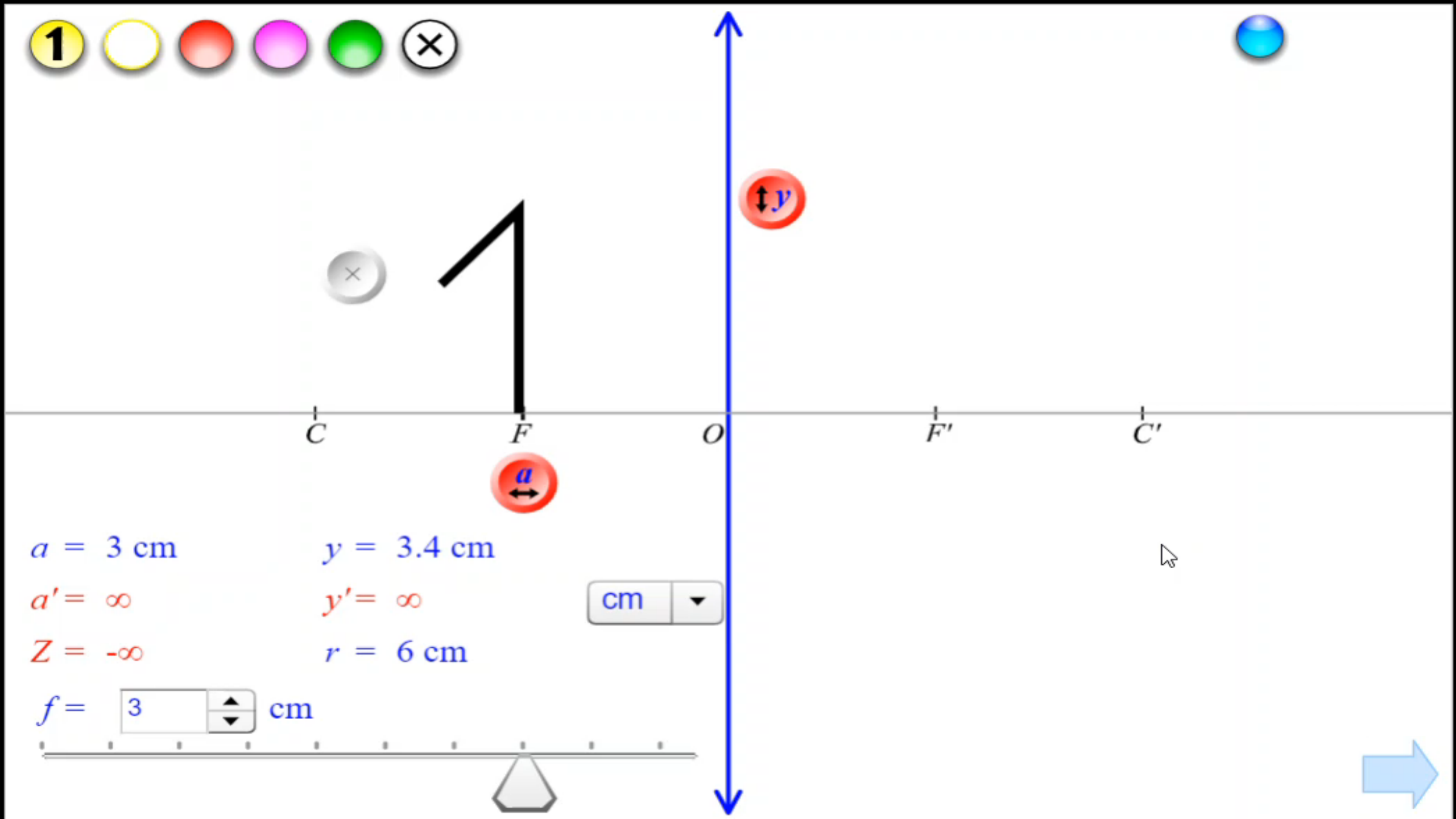
Buyum tasvir o‘lchamining o‘z o‘lchamiga nisbati linzaning chiziqli kattalashtirishi deyiladi.

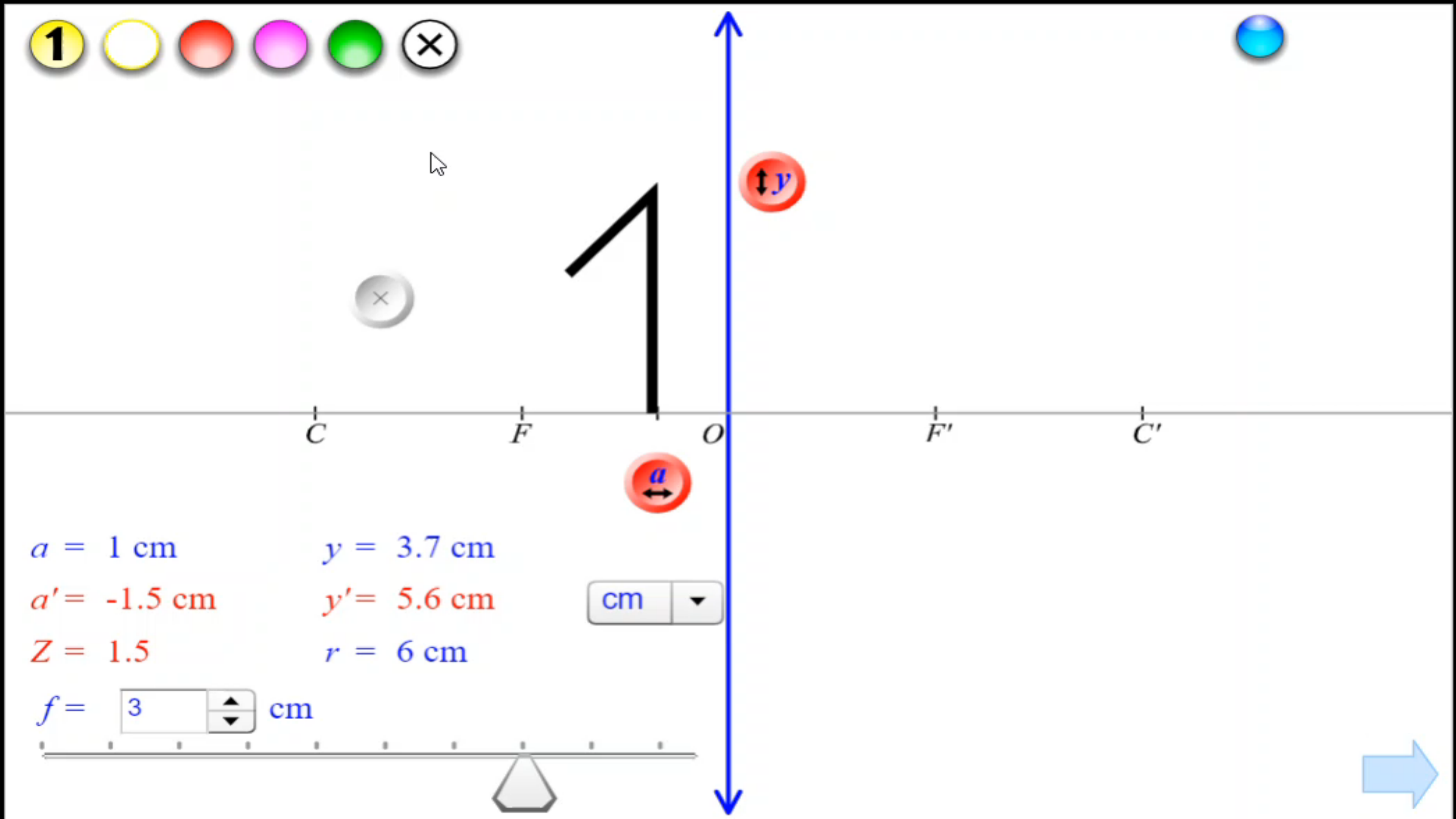
$$K = \frac{H}{h} = \frac{A'B'}{AB} \qquad K = \frac{f}{d}$$


1
y
C
F
O
F'
C'
a
a = 8.7 cm
y = 4.1 cm
a' = 4.6 cm
y' = 2.1 cm
cm
Z = -0.5
r = 6 cm
f =
3
cm

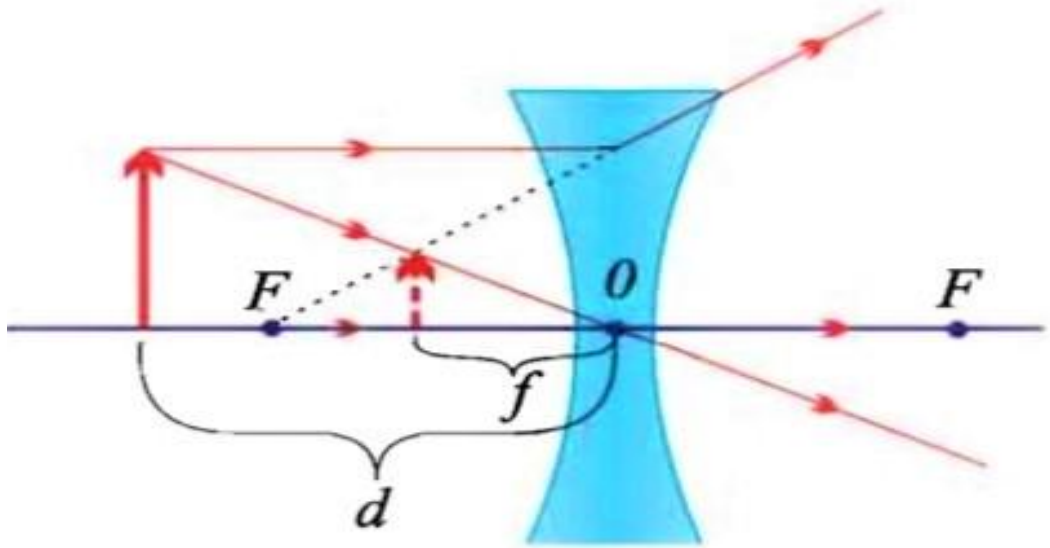

1
y
C
F
O
F'
C'
a
a = 5.9 cm
y = 4.1 cm
a' = 6.1 cm
y' = 4.1 cm
cm
Z = -1
r = 6 cm
f = 3 cm


1
y
C
F
O
F'
C'
a
a = 4.9 cm
y = 3.4 cm
a' = 7.7 cm
y' = 5.4 cm
cm
Z = -1.6
r = 6 cm
f = 3 cm

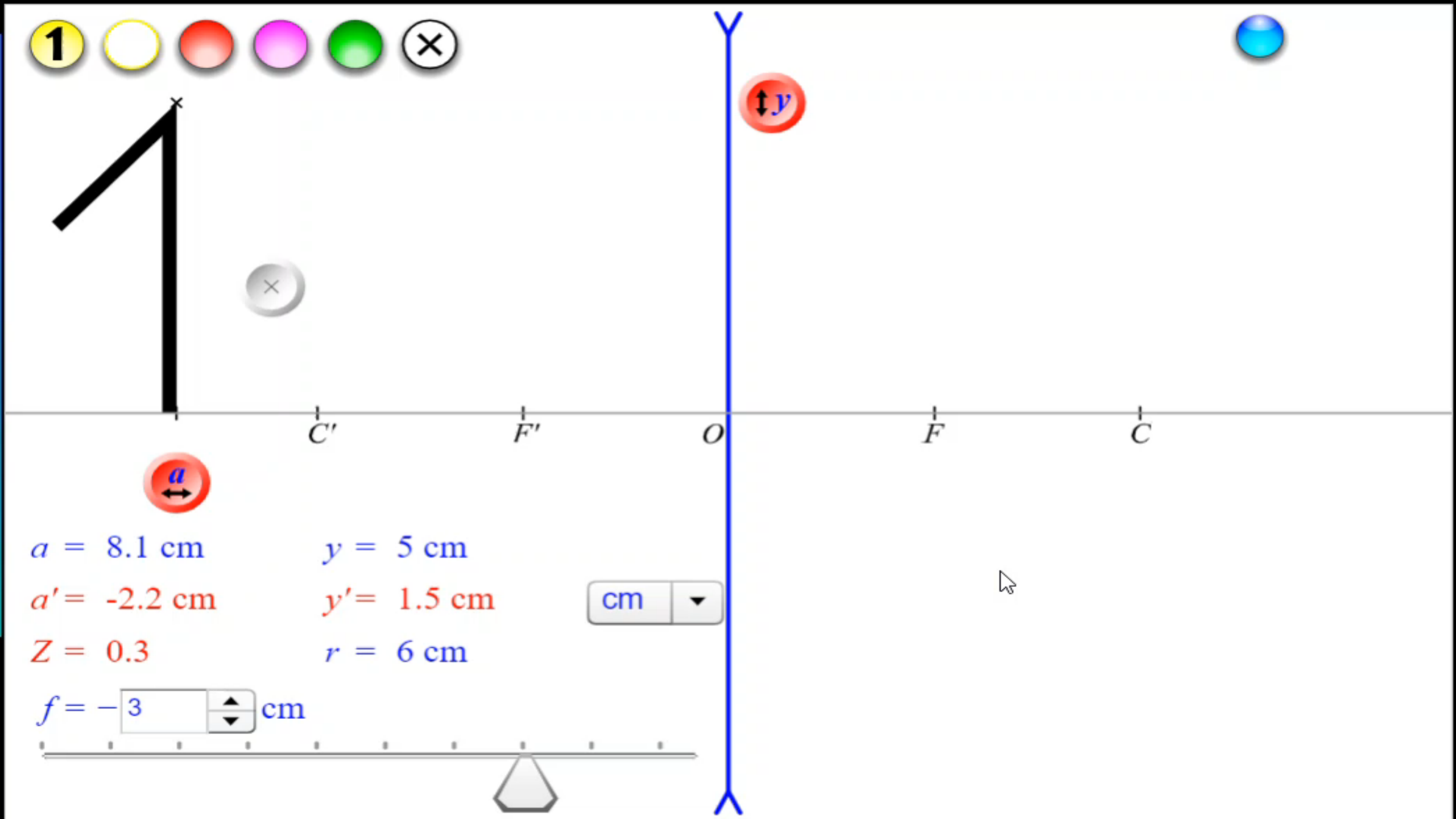

1
y
C
F
O
F'
C'
a
a = 3 cm
y = 3.4 cm
a' = ∞
y' = ∞
cm
Z = -∞
r = 6 cm
f =
3
cm


1
y
C
F
O
F'
C'
a
a = 1 cm
y = 3.7 cm
a' = -1.5 cm
y' = 5.6 cm
cm
Z = 1.5
r = 6 cm
f = 3 cm

# Sochuvchi linzada tasvir yasash




$$-\frac{1}{F}=\frac{1}{d}-\frac{1}{f}$$


1
y
C'
F'
O
F
C
a
a = 8.1 cm
y = 5 cm
a' = -2.2 cm
y' = 1.5 cm
cm
Z = 0.3
r = 6 cm
f = − 3 cm

# Jadvalni to‘ldiring

| Yig‘uvchi linza | | | |
|---|---|---|---|
| $d$ | $f$ | $K$ | Tasvir shakli |
| $d=\infty$ | $f = F$ | $K<1$ (kichraygan) | tasvir haqiqiy, ammo u nuqtasimon va linzaning fokusida joylashadi |
| d>2F | $F < f < 2F$ | $K<1$ (kichraygan) | tasvir haqiqiy, ammo teskari |
| $d = 2F$ | | | |
| $F< d< 2F$ | | | |
| $d < F$ | | | |

# Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish.(147-bet)
2. Jadvalni to‘ldiring.

| Yig‘uvchi linza | | | |
|---|---|---|---|
| $d$ | $f$ | $K$ | Tasvir shakli |
| $d=\infty$ | $f = F$ | $K<1$ (kichraygan) | tasvir haqiqiy, ammo u nuqtasimon va linzaning fokusida joylashadi |
| d>2F | $F < f < 2F$ | $K<1$ (kichraygan) | tasvir haqiqiy, ammo teskari |
| $d = 2F$ | | | |
| $F< d< 2F$ | | | |
| $d < F$ | | | |